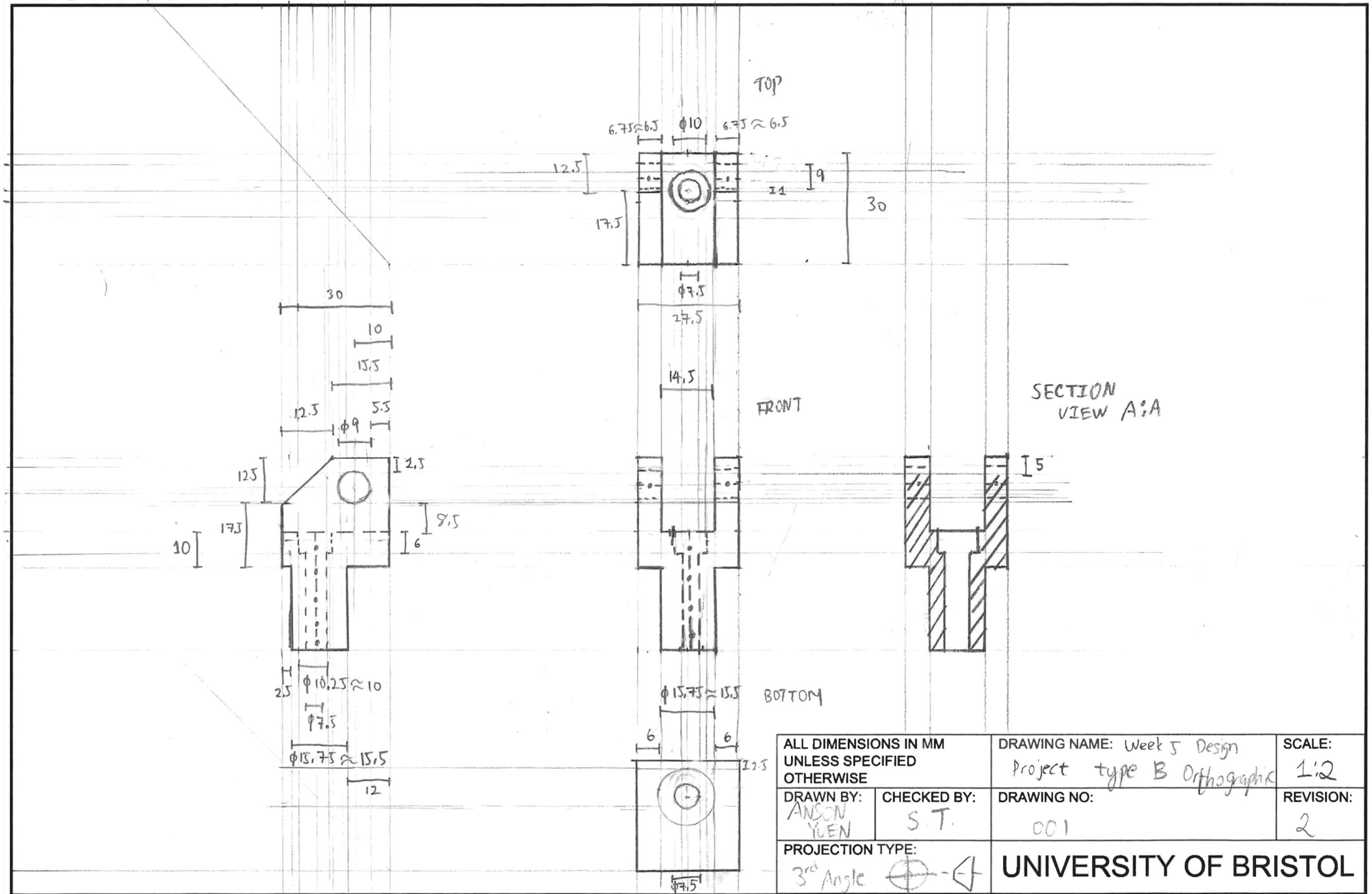
Introduction to Design Practice **Individual Portfolio**

Student Name	**Anson Yuen**	**Date**	**4/12/2024**
Student Number	**2506997**	**Checked**	**P. H.**
ILO	Requirement	Evidence	
1. Communication 1 – Be able to produce basic engineering drawings (hand drawn and/or CAD)	Annex with orthographic and isometric hand drawing and CAD drawings.	See annex A for orthographic and isometric hand drawings (2 side of A3) and annex B for CAD drawings (2 side of A3)	
2. Communication 2 – Be able to explain engineering ideas through words (written and/or verbal)	State that oral presentation was given (will be checked against register) plus annex with artifact study	Gave oral presentation on 26/11/2024 See annex C for artifact study (1 side of A3)	
3. Working safely – Undertake workshop training and take responsibility for evidencing compliance.	Annex with signed off lab induction form	See annex D for signed off lab induction form (1 side of A4)	
4. Design Context 1 – Have an awareness of the UN Sustainable Development Goals and Planetary Boundaries.	No more than 50 words explain which Sustainable Development Goals were addressed in your artefact study and Global Challenges project and which planetary boundaries were considered. Include groups Global challenges project posters	Planetary boundaries: climate change considered as no atmospheric aerosol produced during energy harvest UN SDG goals: SDG 7: affordable and clean energy produced using the solar panels SDG 9, Industry, Innovation and Infrastructure See annex E for Global challenges project posters (4 sides of A3)	
5. Design Context 2 – Be able to describe the local and global context of a project and include this in a design or design review.	No more than 50 words explain the local and global context considered for the Global challenges project	Local context: energy can be harvested in the makers valley community Global context: connects South Africa's IRP project to reduce emissions in line with global climate policy	
6. Design Process 1 – Participate in a conceptual design project and communicate the outcome.	No more than 50 words explaining the design process used on the Global Challenges project plus annex to show Introductory Design Project	Double diamond process used, first we discovered the situation, then decided the specific question, started idea building and looked at case studies. Then began individual brainstorming and had group discussion for down selection of ideas. Finally, final solution was chosen and development continued against requirement before presentation. See annex F for Introductory design project (1 side of A3)	
7. Design Process 2 – Work with a team to produce a design solution.	No more than 50 words explain your main contributions to the group project plus annex with examples of your input	I developed the solar hub solution during the individual brainstorming, which eventually became the final solution. See annex G for evidence of my work on the group projects (1 side of A3)	

Annex A – Orthographic and isometric hand drawing

Drawing border template



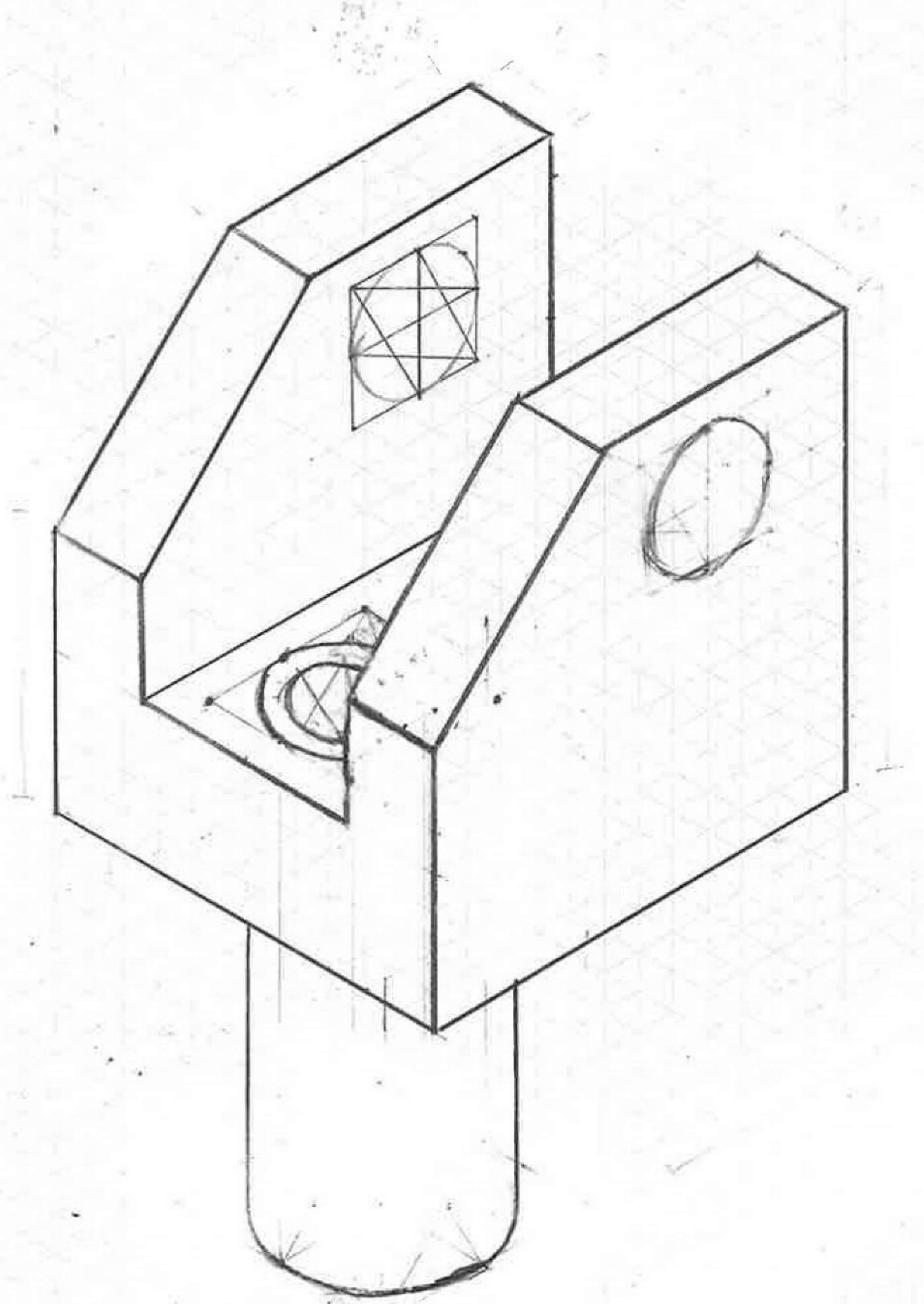
ALL DIMENSIONS IN MM UNLESS SPECIFIED OTHERWISE		DRAWING NAME: Week 5 Design Project type B Orthographic	SCALE: 1:2
DRAWN BY: ANSON YUEN	CHECKED BY: S.T.	DRAWING NO: 001	REVISION: 2
PROJECTION TYPE: 3rd Angle		UNIVERSITY OF BRISTOL	

NOMINAL MAIN BEARING DIAMETER: 31.5 ± 0.025mm

- ALL LINEAR TOLERANCE ± 0.2mm UNLESS STATED OTHERWISE
- ALL ANGULAR TOLERANCE ±1° UNLESS STATED OTHERWISE
- ALL DIMENSIONS

ALL ANGULAR TOLERANCE OF ± 1° UNLESS OTHERWISE STATED
ALL DIMENSION In MM UNO

Drawing border template

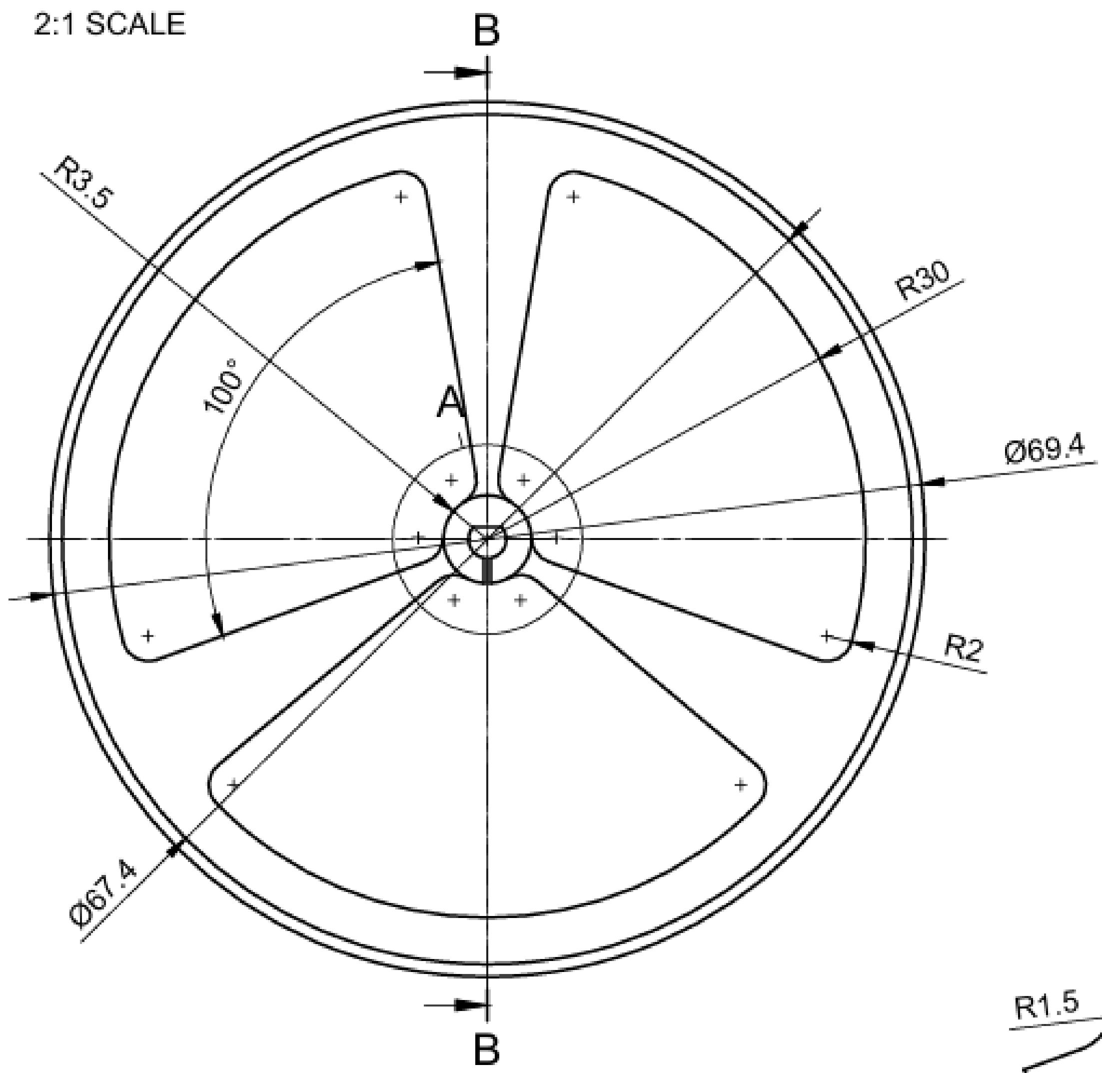


ALL DIMENSIONS IN MM UNLESS SPECIFIED OTHERWISE		DRAWING NAME: WEEK 5 TYPE B DRAWING PROJECT	SCALE: 1:1
DRAWN BY: ANSON YUEN	CHECKED BY: S.T.	DRAWING NO: 002	REVISION: 1
PROJECTION TYPE: ISOMETRIC		UNIVERSITY OF BRISTOL	

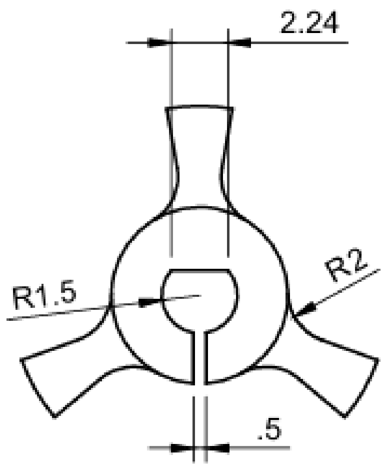
Annex B – CAD drawings

ANGULAR TOLERANCE +/- 1 DEGREE UNLESS NOTED OTHERWISE
LINEAR TOLERANCE +/- 0.2 MM UNLESS NOTED OTHERWISE
ALL DIMENSIONS IN MM UNLESS TOLD OTHERWISE

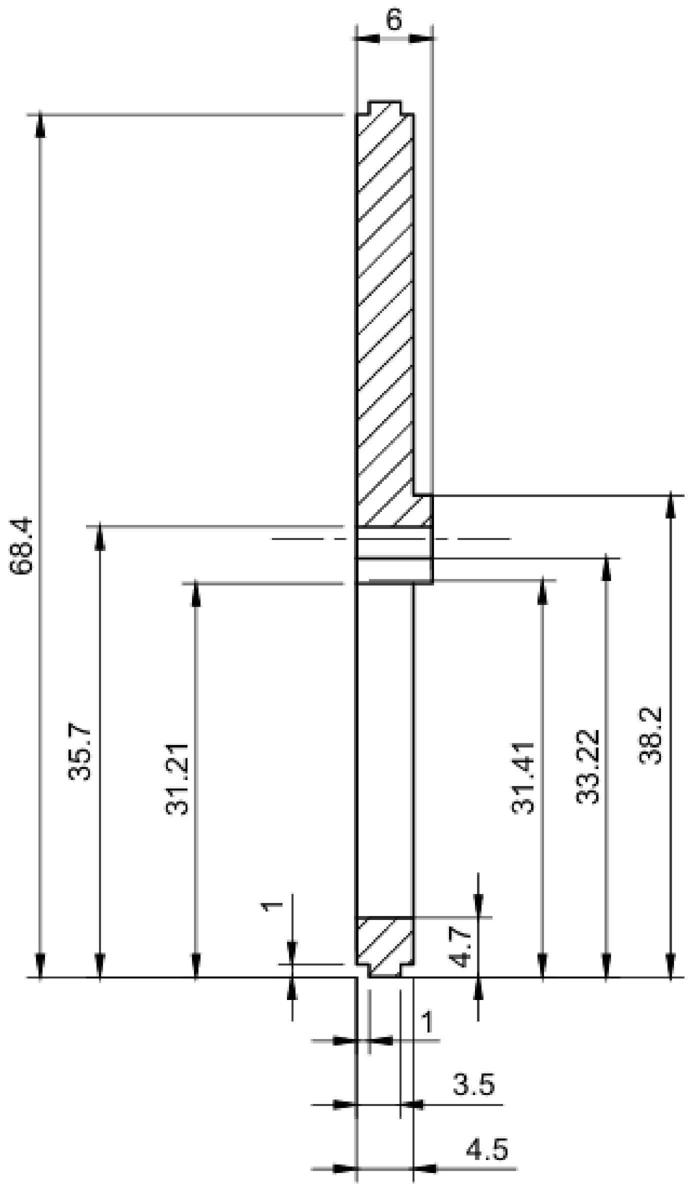
2:1 SCALE

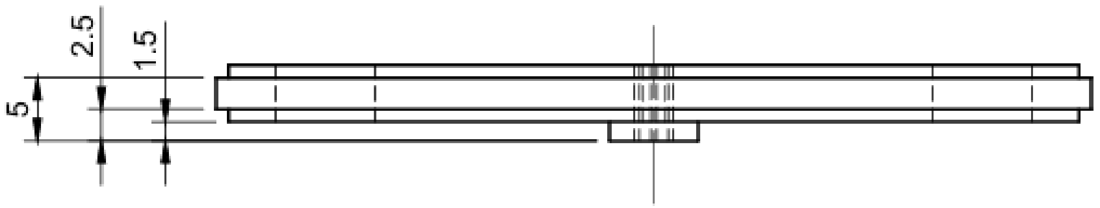


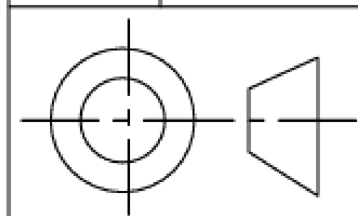
A (4:1)

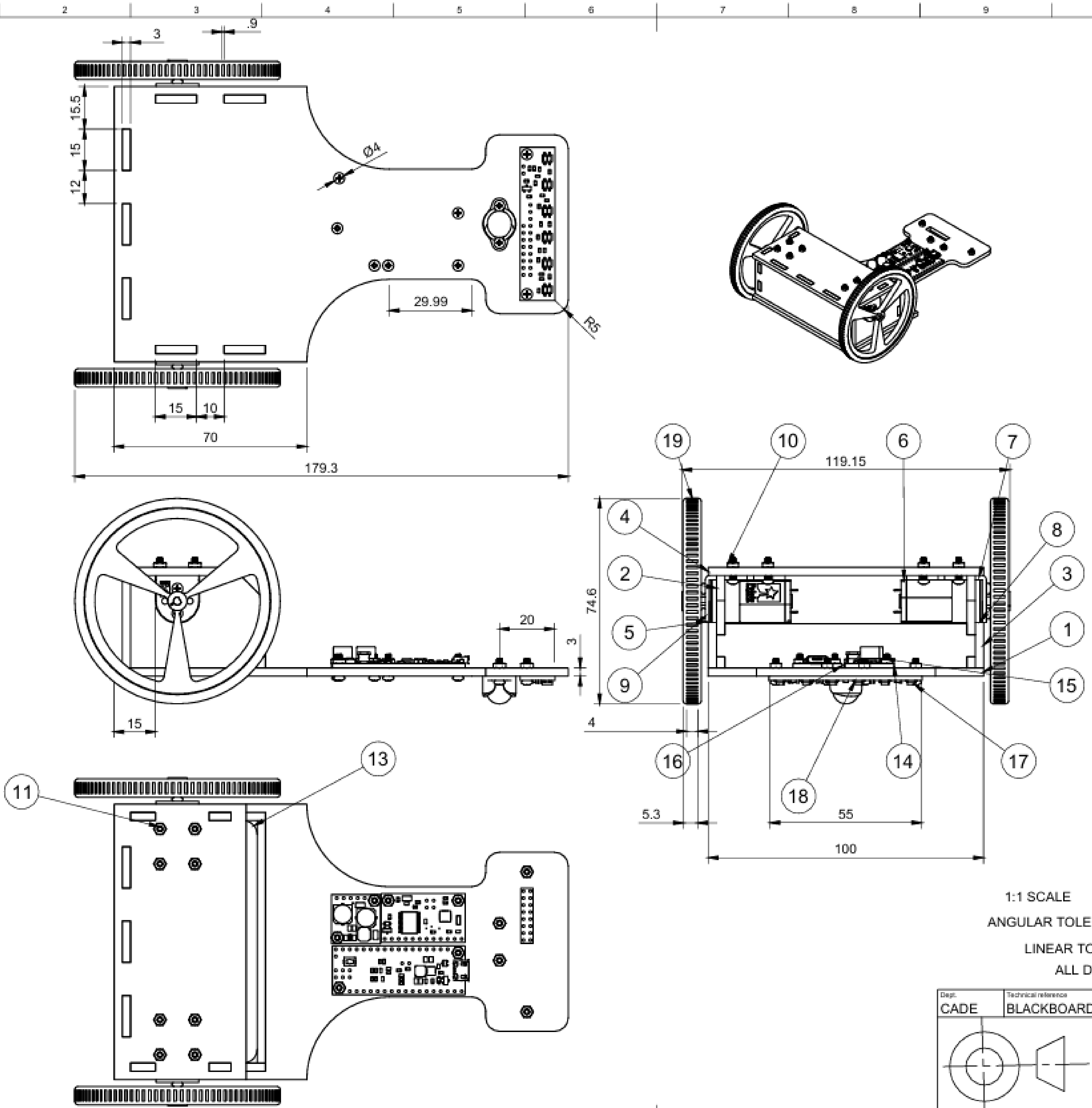


section
view B-B
(2:1)





Dept.	Technical reference	Created by	Approved by
CADE	BLACKBOARD	ANSON YUEN EO23564	matt yang
		Document type: PART DRAWING	Document status: RELEASED
		Title: wheel	DWG No.: 001
			Rev.: 1 · Date of issue: 5/12/2024 · Sheet: 1/1



Parts List			
Item	Qty	Part Number	Material
1	1	Base plate anson	Steel
2	2	Side plate	Steel
3	1	Back plate	Steel
4	1	Top plate	Steel
5	1	92000A011_Passivated 18-8 Stainless Steel Pan Head Phillips Screws	Steel
6	2	GEAR MOTOR	Steel
7	1	GEAR MOTOR BRACKET	Steel
8	3	92000A011_Passivated 18-8 Stainless Steel Pan Head Phillips Screws	Steel
9	1	GEAR MOTOR BRACKET	Steel
10	18	90592A075_Steel Hex Nut	Steel
11	19	92000A015_Passivated 18-8 Stainless Steel Pan Head Phillips Screws	Steel
12	1	Battery cover	Steel
13	1	BATTERY PACK	Steel
14	1	DUAL SERIAL MOTOR CONTROLLER	Steel
15	1	STEP DOWN VOLTAGE REGULATOR	Steel
16	1	PROGRAMMABLE MODULE	Steel
17	1	REFLECTANCE SENSOR ARRAY	Steel
18	1	BALL CASTER	Steel
19	2	Wheel plus tyre	Steel

1:1 SCALE

ANGULAR TOLERANCE +/- 1 DEGREE UNLESS NOTED OTHERWISE

LINEAR TOLERANCE +/- 0.2 MM UNLESS NOTED OTHERWISE

ALL DIMENSIONS IN MM UNLESS NOTED OTHERWISE

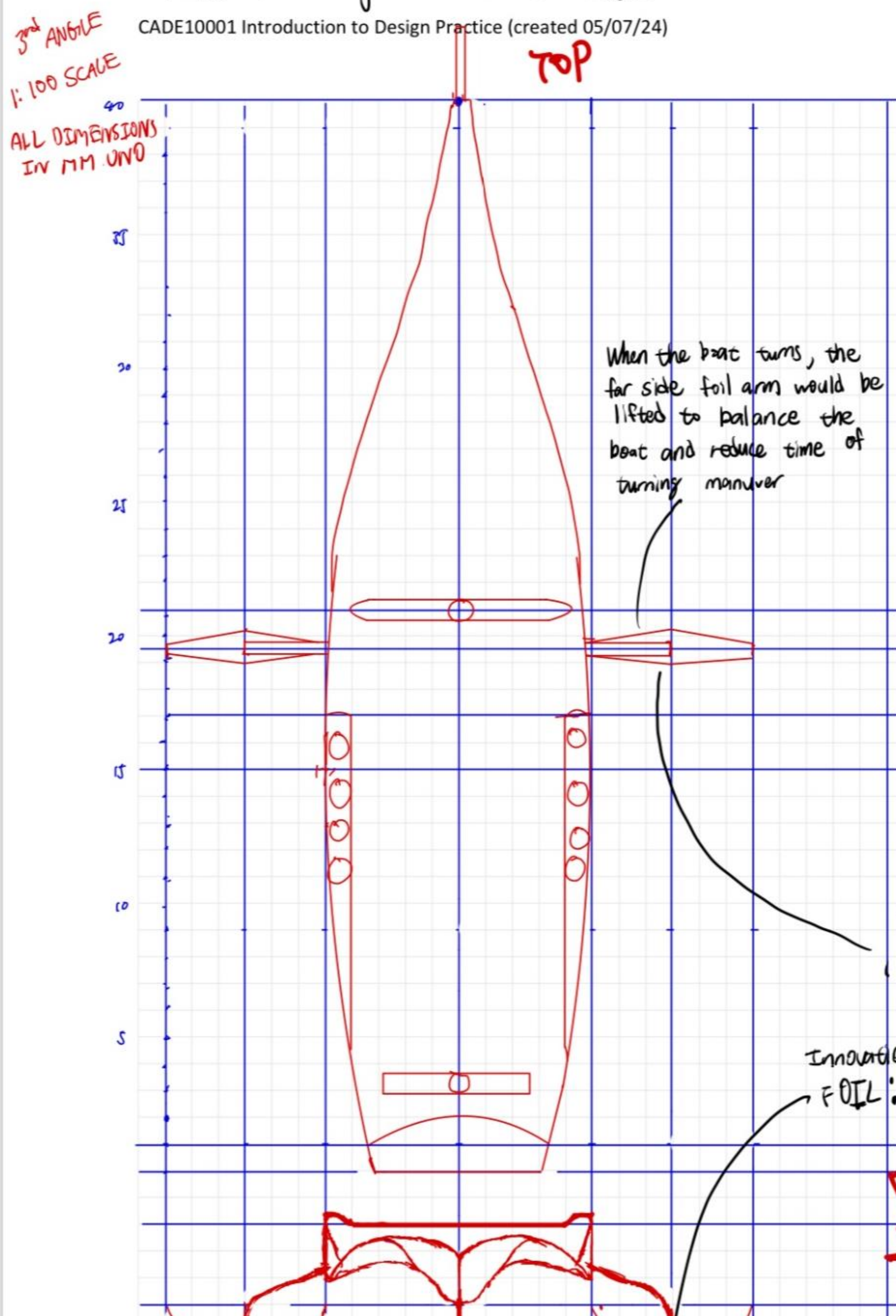
Dept.	Technical reference	Created by	Approved by
CADE	BLACKBOARD	ANSON YUEN 11/29/2024	MATT YANG
		Document type: ASSEMBLY DRAWING	Document status: FINISHED
		Title: LINE FOLLOWING CAR ASSEMBLY	DWG No.: 001
			Rev.: 1; Date of issue: 5/12/2024; Sheet: 1/1

Annex C – Artefact study

(1 side of A3, as submitted in week 8)

Name of design: Hull of Ineos Britannia III

CADE10001 Introduction to Design Practice (created 05/07/24)



- purpose of this design is to challenge team New Zealand in the 2024 America's Cup.
- Design process: - Analyse rules straight after the last tournament
  - Spend 1-2 years to design the boat using CAD, CFD and making small models for wind tunnel testing
  - create a full size prototype ≈ 1 year prior the campaign.
  - get feedback from crew & data of the boat
  - back to the factory for redesign based on feedback
  - final design ≈ 5 months before race begins
- UN Sustainable Development goals considered:
  - SDG 12 (responsible consumption & production):
    - hull completely made of GEN 2 Carbon fibre, which is mostly produced out of recycled carbon fibre, this reduces ≈17 tons of CO_2
    - 3D printing during production of metal (Ti) parts, which produces 8.5 times less waste than the traditional process
- Planetary Boundaries Considered:
  - Climate change
    - As mentioned above, 17 tons of CO_2 saved every 3 years, which contributes to the reduce rate of climate change
- Local Context of design:
  - design & made in the UK, creates lots of job opportunities (engineers, technicans, etc.)
  - Partners with Mercedes F1 during design process, based in Brackley, UK.
    - Gives National Pride, especially when they entered the actual Americas Cup first time in almost 60 years.
- Global Context of design:
  - Partners with foreign companies during design (eg: Suzuki, Grottifredi Maffioli)
  - Stimulates investness, not only local companies but also foreign companies all around the world.
- Innovations included in design:
  - foil arm
  - Bodywork
  - foil

  (see annotations)

Innovation ①

FOIL ARM: connected to the foil, controlled by a hydraulic system with an electric motor, it can be manuvered around a 112° angle.

Innovation ②

FOIL: has a wing structure to generate lift so that the hull is completely out of water during a race. The foil also has flaps to ensure optimum ride height at all time during a race

Annex D – lab induction sign off sheet

CADEIOOOI Introduction to Design Practice

CADE10001 Introduction to Design Practice

Lab induction signoff sheet

Student Name	Anson Yuen	Student Number	2506497
Lab A – Drill Training	QUEENS BLDG L.050 LAB	Date	10/9/2024
Lab B - Soldering	QUEENS BLDG 0.09 ELEC MAIN LAB	Date	27/9/2024

Drill Training	
The student is dressed appropriately	Instructor Initials: EC
The student has brought the right equipment	Instructor Initials: EC
The student has correctly and safely drilled and tapped three number holes	Instructor Initials: EC
The student has behaved appropriately for a lab environment	Instructor Initials: EC
The student has completed their drill training	Instructor Signs: [signature]
Date	Instructor Dates: 20/09/24

Solder Training	
The student is dressed appropriately	Instructor Initials: EC
The student has brought the right equipment	Instructor Initials: EC
The student has correctly soldered the two number specimens	Instructor Initials: EC
The student has behaved appropriately for a lab environment	Instructor Initials: EC
The student has completed their solder training	Instructor Signs: [signature]
Date	Instructor Dates: 27/09/24

Final Induction Signoff	
The student signs here to confirm that: - They have completed the induction training, - They will abide by the laboratory procedures and requirements during future use of laboratories at the University.	Student Signs: AY
Date	Student Dates: 27/09/24

Annex E – Global challenges design posters

(4 sides of A3. Note all group members should use the same posters as evidence)

Engineers Without Borders

MAKERS VALLEY- ENERGY

BRIEF - LOCAL CONTEXT

- Does the power supply match the demand?
  - Over 90% of the population has access to energy through a main grid system, not all daily energy needs are met by this source.
- What is the main source of energy and how expensive is it?
  - South Africa is currently a coal dependent country with over 80% of power currently coming from coal supplied methods. The Integrated Resource plan aims to diversify the power landscape by introducing larger wind and solar components. Due to power source monopoly and general shortage, prices are higher than most residence can afford.
- What measures are currently taken to minimise the issues with the energy system?
  - Currently organised load shedding occurs at a high frequency to prevent total collapse of the system. This however has great impact on education, health and business sectors.

UN SDG's: The goals that this brief considers number & the access to clean affordable energy, it also touches on the responsible consumption and production as well as the sustainable cities and community which are goals 12 and 11 respectively.

Planetary boundaries: the energy crisis currently impacts the CO2 emissions and the atmospheric aerosol loading

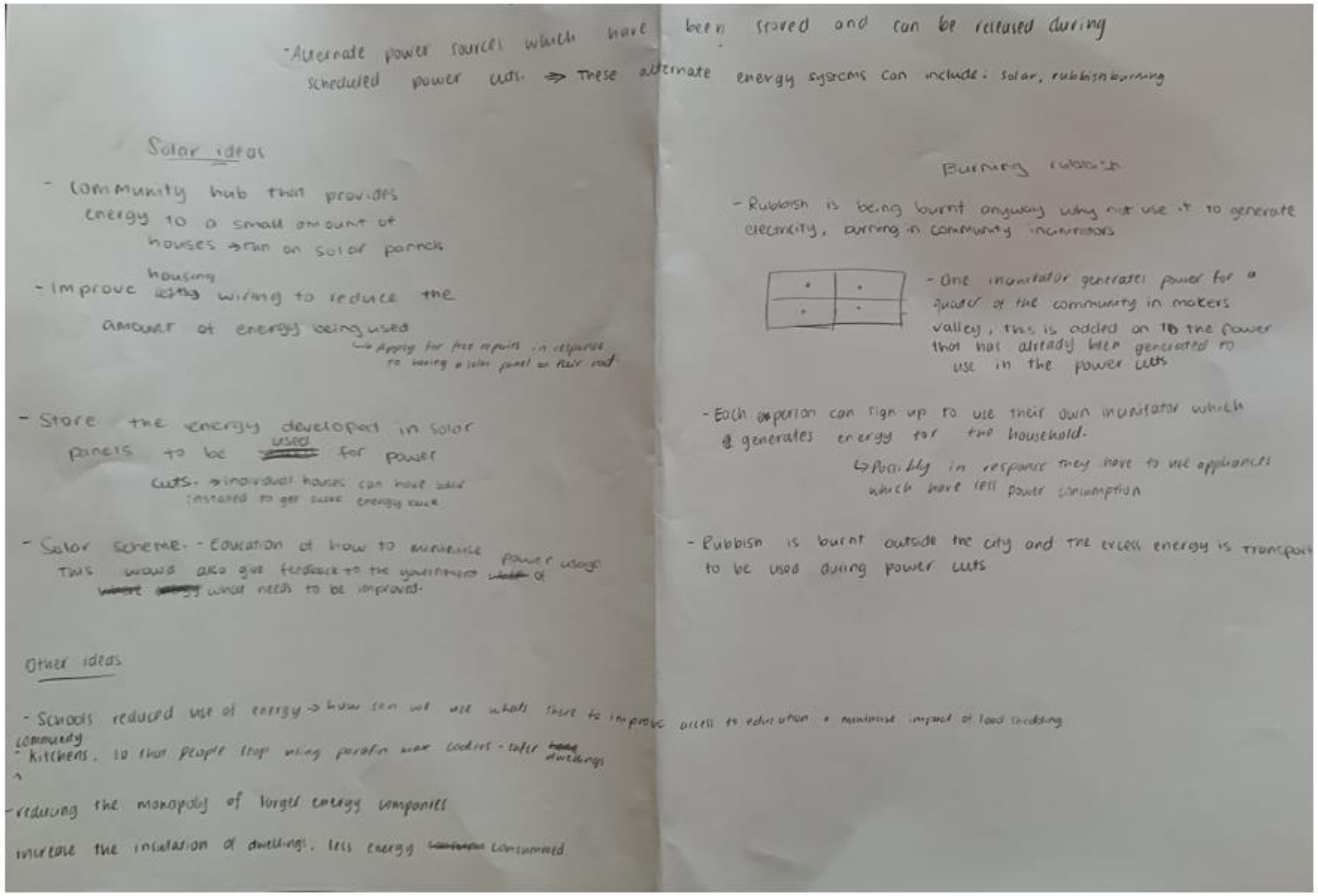
PROBLEM DEFINITION

How can the impact of load shedding be minimised?

Requirements: minimise costs, minimal impact to infrastructure, safety, sufficient energy production, environmental impact, long term impacts

INTIAL SOLUTION BRAINSTORM

- Waste-to-energy incinerators
- Solar Fields
- Solar Panels on Roofs
- HEP Generator
- Safety Rewiring
- Introduce other power plants



- Alternate power sources which have been stored and can be released during scheduled power cuts. ⇒ These alternate energy systems can include: solar, rubbish burning

Solar ideas

- Community hub that provides energy to a small amount of houses → run on solar panels
- Improve ~~existing~~ housing wiring to reduce the amount of energy being used
  ↳ Apply for free repairs in response to having a solar panel on their roof
- Store the energy developed in solar panels to be used for power cuts. → individual houses can have solar installed to get some energy back
- Solar scheme. - Education of how to minimise power usage. This would also give feedback to the government of what needs to be improved.

Other ideas

- Schools reduced use of energy → how can we use what's there to improve access to education + minimise impact of load shedding
- community kitchens, so that people stop using parafin war cookers - safer dwellings
- reducing the monopoly of larger energy companies
- increase the insulation of dwellings, less energy consumed

Burning rubbish

- Rubbish is being burnt anyway why not use it to generate electricity, burning in community incinerators
- One incinerator generates power for a quarter of the community in makers valley, this is added on to the power that has already been generated to use in the power cuts
- Each person can sign up to use their own incinerator which generates energy for the household.
  ↳ Possibly in response they have to use appliances which have less power consumption
- Rubbish is burnt outside the city and the excess energy is transport to be used during power cuts

DEVELOPED SOLUTION 1

Waste collection service, and burn in incinerators, to provide an alternate energy source to nearby blocks, during hours of load shedding. Working with local waste management companies to gather waste.

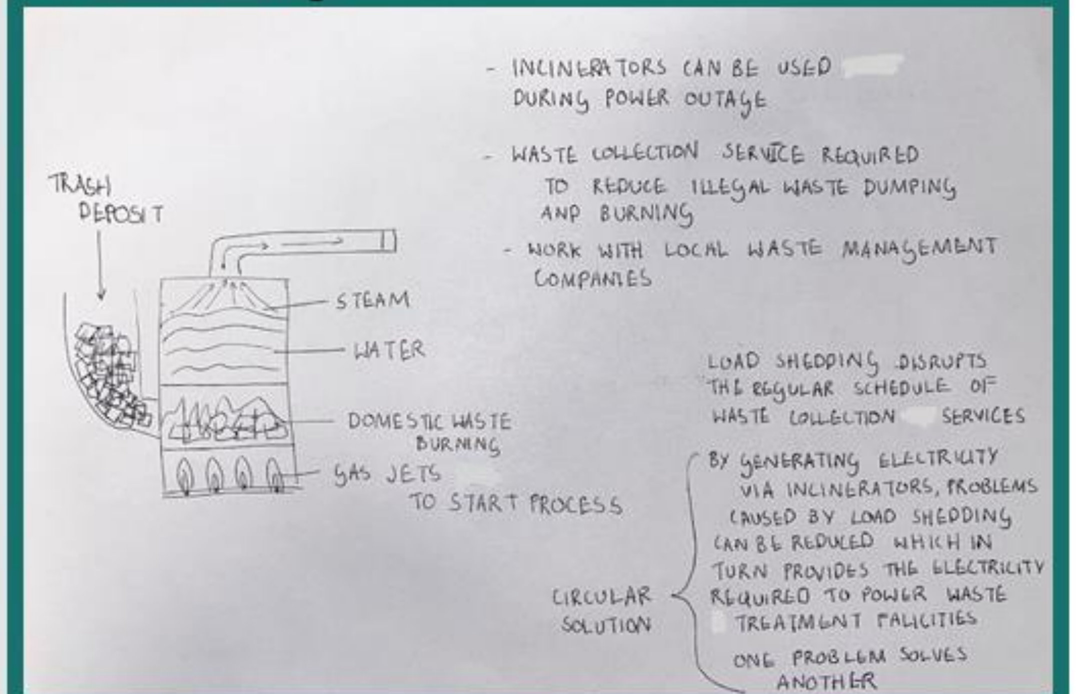
LOCAL CONTEXT: trash is already being burnt for heat and waste disposal within these communities

GLOBAL CONTEXT: globally main developed countries ship garbage to be burnt for energy to be returned

UN SDG:

SDG - 15, Life on land, by preventing degradation of land.

PLANTARY BOUNDARIES: does not consider the planetary boundary of climate change (atmospheric aerosol loading).



DEVELOPED SOLUTION 2

Solar panels are placed on the top of the building or on empty ground, they are connected to a battery pack to store the energy.

LOCAL CONTEXT: energy can be harvested in the makers valley community for the community

GLOBAL CONTEXT: connects to South Africa's IRP project to reduce emissions in line with global climate policy

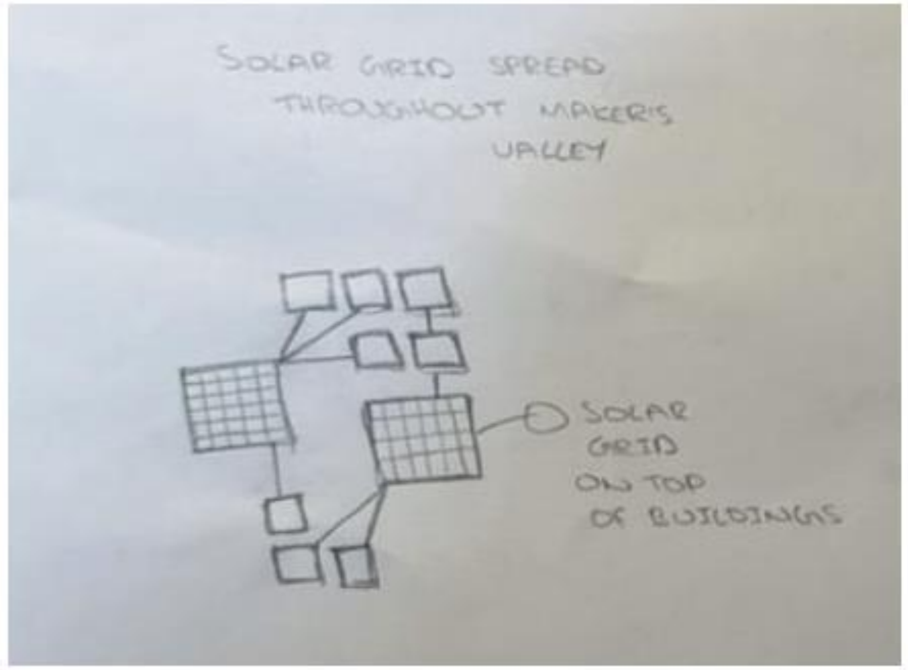
UN SDG:

SDG - 7 - affordable and clean energy

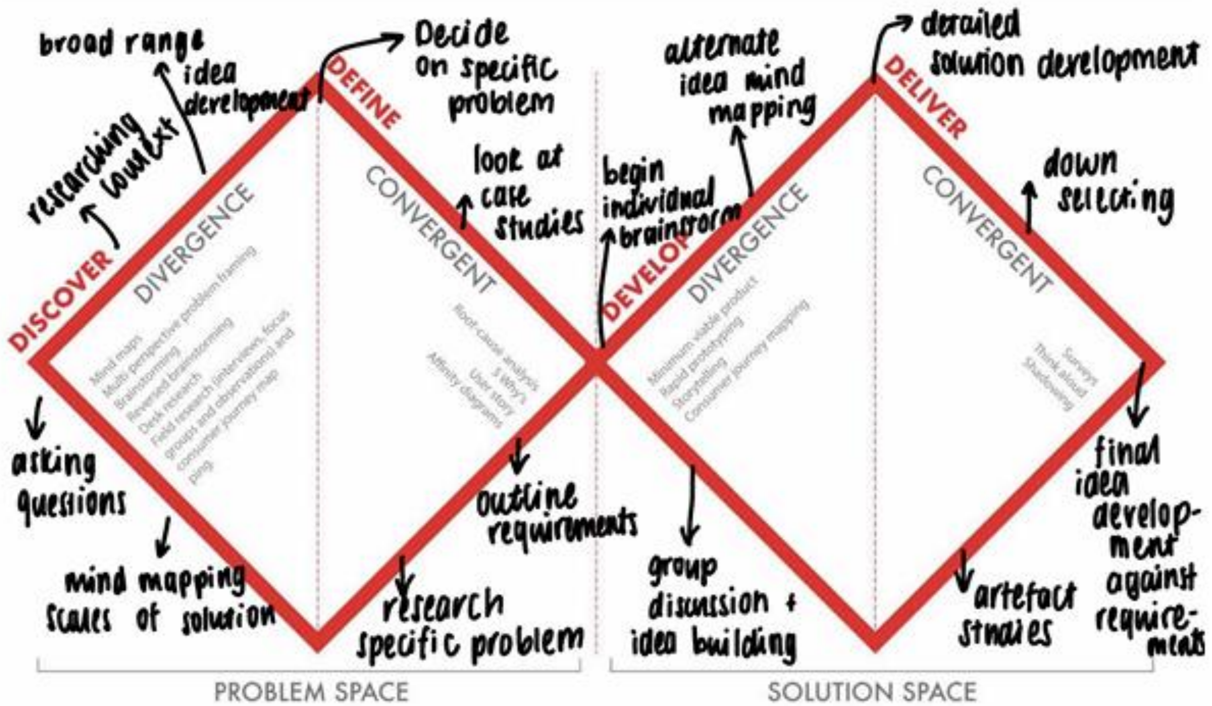
SDG - 9 - Industry, Innovation and infrastructure.

PLANTARY BOUNDARIES:

Planetary boundary of climate change considered as there is no atmospheric aerosol

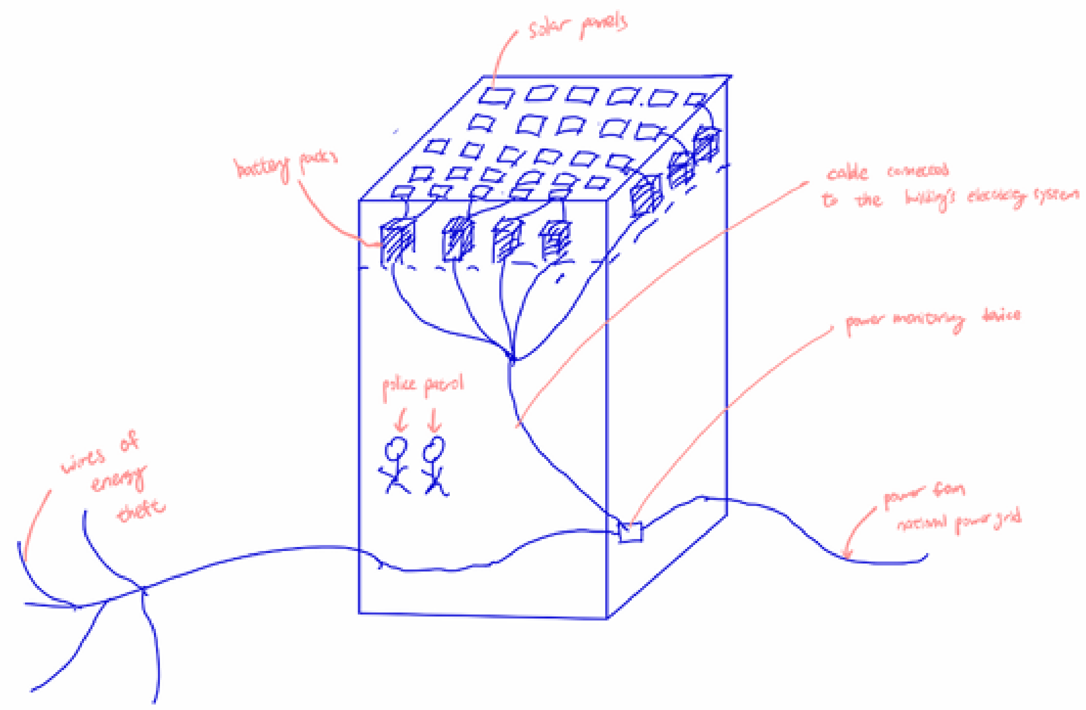


DESIGN PROCESS



DOWN SELECTING

Solutions	Criteria						
	Energy saved/generated	Cost	Energy efficiency	Environmental impact	Size	Long term impact	Total
solar panels	0	0	0	0	0	0	0
trash incinerator	2	-3	0	-3	-2	-1	-7
wire safety	-3	3			1	-2	-1
hydroelectric	-1	-2	2	-1	-1	0	-3



FINAL SOLUTION - SOLAR

Our aim is not to eliminate load shedding but to give alternate solutions that provide energy while load shedding is occurring. This solution most effectively meets the requirements set out in the brief.

Advantage:

1. Promotes sustainability and green energy in South Africa (one of the least energy sustainable country in the world)
2. Solves the problem of load shedding (at its best)
3. No new buildings are built, relatively cheaper
4. Solar panel is one of the cheapest out of the renewable energy
5. There are buildings with solar panels in johannesburg so we can foresee the local problems
6. Relies on current wire system so no need to build a new one

Disadvantages:

1. The investor probably won't get their money back so it probably has to be a charity to do it
2. Solar panels are still very expensive and require constant maintenance
3. Solar panels relies on the suns, so the energy input is not constant and stable, eg. no energy input at night or cloudy days
4. This system is in a way encouraging energy theft
5. Extra maintenance for the wiring to make it safer

Annex F – Initial design project poster

(1 side of A3, as submitted in week 5. Note all group members should use the same poster as evidence)

CADE10001 Introduction to Design Pra...

Drawing border template

Martian Resources at our Disposal:

- Soil
- Sun
- Geothermal
- Minerals
- Batteries
- Manufacturing equipment
- Water
- Seeds
- Fertiliser
- Spacesuits
- Construction materials

: Resources we are using

Essentials for Society on Mars:

- Food
- Water
- Shelter
- Medical supplies
- Toolkits
- Communication systems
- Power Sources
- Air
- Clothes
- Cooking equipment
- Waste management
- Pressure System
- Exercise
- Hygiene
- Entertainment

: Essentials we're adressing

Questions for Society and its Success

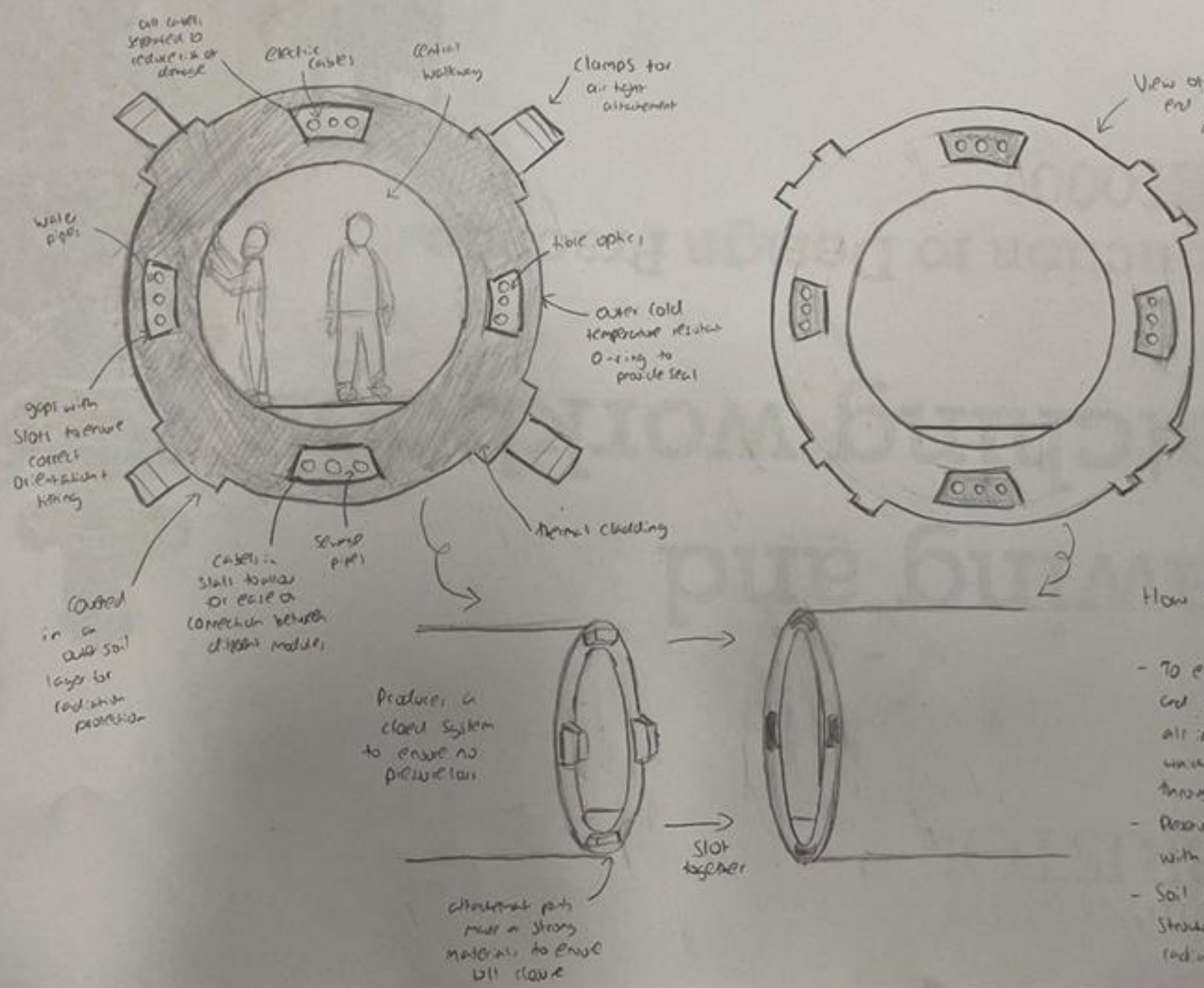
1. How are we going to produce sufficient power?
2. How are we going to process waste?
3. How are we going to connect modules for easy expansion?
4. How do we replace broken parts?

: Question that we chose

Why we chose our Question:

Table 29

- Mars will be an expanding community, so the habitat needs to grow as well.
- It is also possible that modules are damaged or need moving and a simple attachment system allows this to happen.
- How to maintain pressure systems and run water, power and internet from source to user.



How Did We Make Our Final Choice?

- To ensure constant power and good connections between all important substances which need to be transported throughout entire habitat
- Preserve original orientation with slots + clamps
- Soil covered to ensure structure is protected from radiation

Annex G – Evidence of individual input for group projects

(1 side of A3. Can included sketches, drawings, tables, text, diagrams, and photos that capture your **individual** contribution to the group projects)

Solar panel system (Anson):

- Will be built on site where energy theft occurs
- **Our aim is not to eliminate energy theft but to give them a more stable electricity supply because we understand that they probably have a reason behind their actions**
- Solar panels are placed on the top of the building or on empty ground, they are connected to a battery pack to store the energy. This might however require a larger space.
- The battery packs would be connected to the empty buildings electricity system
- A device would be used to monitor the situation of the power supply by detecting the voltage and current values at all times, and an alert would be sent out when a problem appears (load shedding)
- When warning is detected the alternate power system would be triggered and the electricity in the battery would kick in
- The rate of energy supply would be at the lowest point in the last 24 hours
- The building would have armed police patrol 24/7 and an emergency line to the nearest police station to ensure no physical theft happens on site (solar panels)
- Theft wires would be monitored and no more extra wires are allowed to connect to the system, so ensure no extra speculators, and enough power supply

Advantage:

1. Promotes sustainability and green energy in South Africa (one of the least energy sustainable country in the world)
2. Solves the problem of load shedding (at its best)
3. No new buildings are built, relatively cheaper
4. Solar panel is one of the cheapest out of the renewable energy
5. There are buildings with solar panels in johannesburg so we can foresee the local problems
6. Relies on current wire system so no need to build a new one

Disadvantages:

1. The investor probably won't get their money back so it probably has to be a charity to do it
2. Solar panels are still very expensive and require constant maintenance
3. Solar panels relies on the suns, so the energy input is not constant and stable, eg. no energy input at night or cloudy days
4. This system is in a way encouraging energy theft
5. Extra maintenance for the wiring to make it safer

Speech

Final Solution: Anson

In the end we chose the solar hub idea where the current and voltage of power supply would be monitored at all time in a building where energy theft occurs, and when a loading shedding pattern is detected the monitoring device sends an alert to the alternate power system and the battery pack kicks in to maintain power supply. The rate of it would be the lowest point in the last 24 hrs. Also the theft wires would be monitored to make sure no more wires connect to the system to ensure enough power supply and police patrol at all time to ensure no physical theft. This design would be able to minimise the affect of load shedding while not producing extra pollutant, with a relatively cheaper price.